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Cerebral hydatidosis: Multicentre study

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Background and aims

Hydatidosis is still prevalent in our country and represents a real public health problem. Its localization in the brain represents 2% of all hydatid localizations. Aims: To study the epidemiological, clinical, radiological, histological, therapeutic and evolutionary parameters of cerebral hydatid cysts; evaluate the vital and functional prognosis and highlight the statistical correlations with the different parameters studied.

Methods

This was a multicenter retrospective study on cases of cerebral hydatid cysts operated on in the neurosurgery services of the six hospitals in Tunisia over a period of 20 years.

Results

87 patients were included. The clinical picture was not specific. Imaging allowed the diagnosis in 83 cases with a sub-tentorial localization in 9.2%. Cerebral hydatid cysts were classified as type I in 73 cases, types II and III in 6 cases and type V in 3 cases. They were classified as secondary in 15 cases. Four patients were operated on in emergency, 51 patients in delayed emergency and 32 patients were scheduled for cold surgery. A favorable short-term outcome was noted in 71.3% of cases. Ten patients had a recurrence. A favorable long-term outcome was noted in patients with lower median ages and shorter delays in treatment. The mortality rate was 4.8% and was statistically significantly related to age and consciousness disorders.

Conclusions

Hydatidosis is still endemic and continues to pose serious health and economic problems in our country. This requires the implementation of a national program to fight against this disease in order to eradicate it.

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Clinico-radiological assessment and therapeutic management of intracranial suppurations

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Background and aims

Background: Intracranial suppurations (ICSs) are purulent collections of infectious origin. Many therapeutic attitudes have been described. Management has not really been the subject of consensus. Aim: Study the epidemiological, clinical, and paraclinical features of ICSs and assess the clinical, radiological and postoperative outcomes according to the different therapeutic modalities.

Methods

Patients and methods: Our retrospective, descriptive study including 43 cases of ICSs treated in the Neurosurgery department of the UHC

Habib Bourguiba and Traumatology and Major Burns Center of Ben Arous-Tunisia, over a period of 7 years from 2012 to 2018. All epidemiological, etiological, clinical, radiological, biological, therapeutic and evolutionary parameters were studied.

Results

The mean patients'age was 32.35 years with a sex ratio of 4.37. Locoregional infections accounted for 60.46% of aetiologies. Clinical signs found were infectious syndrome in 79.07% and intracranial hypertension in 62.70%. Neuroimaging made diagnosis in 30 cases of abscess and 13 cases of empyema. Predilection site was supratentorial in 90% of abscess and in 92.30% of empyema. Streptococci were isolated in 75% of abscess and in 4 cases of empyema. All patients received broad-spectrum antibiotics. Abscess aspiration was performed in 20 cases and excision in 17 cases. Evacuation of empyema by trepanation was performed in 7 patients and through craniotomy in 7 cases. Outcomes were favorable in 23 cases of abscess and 11 cases of empyema.

Conclusions

ICSs are a topical pathology in various countries where management is multidisciplinary. Outcomes are favorable in more than 80%, improved by early treatment. Clinico-radiological monitoring during follow-up remain essential.

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Bilateral optic disc edema by neurosyphilis

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Background and aims

Optic disc edema is closely associated with central nervous system disorders, such as intracranial hypertension, anterior ischemic optic neuropathy and infections, and represents a diagnostic challenge in immunocompromised patients.

Methods

We reported a case of a patient presenting bilateral optic disc edema but no other neurological signs or symptoms, which was defined as secondary to neurosyphilis.

Results

A 59-year-old male patient with HIV and an undetectable viral load presented with worsening vision in the left eye; an optic disc edema was found on the left at the time. Two months later, he sought emergency care due to blurred vision in the right eye, with bilateral optic nerve head edema being identified. Laboratory tests showed positive Venereal Disease Research Laboratory (VDRL) in peripheral blood (1/256), but negative VDRL in cerebrospinal fluid (CSF). However, the analysis of the CSF showed findings compatible with the diagnosis of neurosyphilis: a slight increase in cellularity (5 cells) and protein levels at the upper limit of normality (40 mg/dL). Serology for CMV was also requested, with positive IgG and negative IgM results. Magnetic resonance imaging and angiotomography of the skull revealed no abnormalities. He was treated with crystalline penicillin

for 14 days, after which the optic disc edema and the visual complaints resolved completely.

Conclusions

Syphilis is a chronic bacterial infection capable of causing several neurological manifestations, especially in patients with HIV. Therefore, it should be considered in the differential diagnoses of this population to guarantee early treatment and avoid greater neurological and visual damage.

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Neurobrucellosis and vitamin B12 deficiency: Coincidence or common physiopathological mechanism?

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Background and aims

Brucellosis is a rare zoonotic infection, endemic in Mediterranean region. It could involve multiple systems. Neurobrucellosis is an uncommon complication occurring at any stage of the disease with heterogeneous clinical presentations.

Methods

Case report.

Results

Case report We report the case of a 61-year-old man who presented since 2 years progressive tingling sensations in the lower legs followed by motor deficit and walking difficulties. He had an approximate 1-year history of bilateral hearing loss. There was no history of cephalalgia. There was a history consumption of unpasteurized fresh milk. His neurological examination demonstrated spastic paraparesis with thigh pain and hypopallesthesia in lower legs. Cerebral and spinal magnetic resonance images with contrast showed signs of a leptomeningitis with meninges thickening. Cerebrospinal fluid analysis showed meningitis (10 cells) with hyperproteinorachia at 0.57 g/L and hypoglucorachia (2.95 mmol/L; $R = 0,392$). Rose Bengal test was positive in the serum. Wright test was positive with a titer of 640. Vitamin B12 dosage was 130 pg/mL. There was neither anemia nor macrocytosis. ENMG showed axonal neuropathy in lower limbs.

Conclusions

Clinico-radiological symptoms of neurobrucellosis are oftenly atypical mimicking other neurological diseases. Chronic meningitis is the most frequent clinical presentation. Spastic paraparesis is rarely reported, oftenly not understood. Association of neurobrucellosis and vitamin B12 deficiency is reported in one case in the literature in which physiopathological mechanism was not elucidated. In endemic areas, neurobrucellosis should be considered for any unexplained neurological symptoms.

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Central nervous system tuberculosis causing motor tractopathy-a case report

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Background and aims

A 46-year-old previously well farmer presented with fever, headache, diplopia, gait difficulty, and constitutional symptoms for two weeks. Examination showed bilateral asymmetric spastic paraparesis with left-sided abducens and lower motor-type facial nerve palsy. There was no sensory system or cerebellar involvement. The inflammatory markers were raised. CSF analysis showed lymphocytosis, and elevated protein with a significant sugar drop. CSF TB GeneXpert and cultures were repeatedly negative. HIV, VDRL, Melioidosis antibodies, and the rest of the septic screen were negative. NMO and MOG antibodies were negative. Pulmonary tuberculosis was ruled out. MRI Brain showed basal meningeal enhancement and bilateral pyramidal tract T2 hyperintensities with contrast enhancement. MRI spine with contrast showed a long segment signal change in lateral corticospinal tracts extending down to the upper part of the thoracic cord.

